

### NUCLEAR GIANT

*The Consolidated B-36 was powered by an array of pusher propellers and turbojets. It was so large that the crew of 15 moved from the front to the back of the bomber by pulling themselves along a tunnel on a trolley, and slept in bunks when off duty.*

YB-49, as America's strategic nuclear bomber. But after much discussion, the air force instead opted for the Consolidated B-36, an aircraft that was remarkable above all for its sheer size – one pilot said it was “like flying an apartment house.” The B-36's wingspan of 230ft (70m) was more than double that of a World War II B-17 – in fact considerably bigger than today's Boeing 747. A physical monument of the transition to the jet age, it had six pusher propellers plus four turbojets, an array of power that made the ground shake as it flew overhead. Deployed alongside the B-47, the B-36 provided a stopgap solution until the Boeing B-52 took over as the hub of America's nuclear bomber force in the second half of the 1950s.

### Soviet counterparts

To the dismay of Americans, the Soviet Union seemed able, time and again, to match the progress of Western technology or even surpass it. In retrospect, it is obvious that the American perception of Soviet strength was frequently exaggerated, with misleading intelligence reports fueling ill-considered paranoia. Never was this more true than during the “bomber gap” crisis of the mid-1950s, when many Americans came to believe that the Soviet Union was well ahead of the United States in the creation of a strategic nuclear bomber force.

In fact, the Soviets found it even more difficult than the United States to create a credible intercontinental nuclear bomber. When the Soviet Union exploded its first atomic bomb in 1949, it could only have been used against the United States on a suicide mission, since the Soviets' Tupolev Tu-4 bomber would have had the range for a one-way journey only. It was not until the Tu-95 entered service in 1956 that the Soviets had a bomber of true intercontinental range – and this was a turboprop aircraft.

struts under the wing. It was considered fast enough to penetrate Soviet air defenses, cruising at around 560mph (890kph), but it lacked intercontinental range – therefore it would either have to be based on the territory of America's allies nearer the Soviet border, or rely heavily on in-flight refueling.

For a true intercontinental bomber, the USAF looked at one point as if it would turn to America's most innovative designer, Jack Northrop. Since the early 1940s, Northrop had been working on the XB-35, a propeller-driven flying-wing bomber with neither fuselage nor tail. He proposed a jet-powered version, the

### SOVIET BEAR

*Introduced into service in 1956, the Tupolev Tu-95 Bear turboprop bomber gave the Soviet Union, for the first time, the capacity to deliver a nuclear strike against the United States.*



### DETERRENT FORCE

*A still from a 1949 documentary, Target: Peace, shows a crewman's view of a fleet of B-36s in operation. After it was withdrawn from service, the B-36 was nicknamed “Peacemaker” to stress that its purpose was to prevent war through deterrence.*

### Interceptors

Still, the threat of aircraft destroying a city with a single bomb understandably gave both sides in the Cold War an urgent desire to develop interceptor aircraft. With the speed of bomber aircraft increasing constantly, it was estimated that from the time an enemy incursion was identified by radar, an interceptor would have little more than 10 minutes to reach and destroy its target.

The specialized interceptors designed in the 1950s were pilot-operated missile-platforms maximized for speed in a straight line and rate of climb, with electronic systems on board linked directly to ground controllers. The F-106, dubbed the “Ultimate Interceptor,” flew at over Mach 2, could climb to almost 40,000ft (12,000m) in a minute, and, having no guns, depended entirely on its air-to-air missiles.

Aircraft with this level of performance and complexity of electronic systems were massively expensive. An F-106 cost more than 10 times as much as an F-86 Sabre to build, and its operating costs were similarly astronomical. For countries trying to keep up with the United States and the Soviet Union, the economic demands were daunting. Nonetheless, the French, eager to assert

